# The Falsifiable Clinic

*Deterministic-first clinical intelligence as sovereign infrastructure: from the OCCR specification to a conformance class, an equity-parity covenant, and a manufacturing corridor for the majority world.*

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Thesis — The defining health-infrastructure decision of the next decade is not which artificial intelligence to buy; it is which epistemology to procure. Stochastic clinical AI cannot be governed: its accuracy is population-bound, its failures are detected post-market, and its logic cannot be inspected. OCCR's deterministic-first architecture converts clinical AI from a product a nation must trust into infrastructure a nation can audit — with equity as a measured covenant (EPI < 0.05), falsifiability as a conformance test, and sovereignty as a manufacturing specification. The majority world is not the last market for this architecture. It is the proving ground from which it will generalize.

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C0 INFORMATIONAL · REGISTERED RESEARCH PROGRAMME — INSTRUMENTS FROZEN IN ADVANCE;
PROSECUTION-SENSITIVE DETAIL WITHHELD PENDING IP REVIEW

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| This edition | This public edition omits the internal commissioning, vehicle, intellectual-property, and funding apparatus of the original blueprint. The instrument, its hypotheses, and its kill criteria are complete. |

## How to read this edition

Parts I–III establish the crisis and prove the territory is unoccupied. Part IV extends the parent specification into a formal conformance and financing framework. Part V is the empirical programme — pre-registered, with explicit rejection criteria. Part VI specifies the instrument suite. Part VII states the risk register and the kill criteria the programme binds itself to. Every claim is engineered to be falsifiable; where evidence is thin, the document says so and proposes the test, not the conclusion. The parent work's own closing line is the discipline this blueprint inherits: the architecture is not a product; it is infrastructure.[1]

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ORIENTATION

# Executive Summary

**Medicine's reasoning layer is becoming national infrastructure — and it is the only infrastructure class the world procures without an audit discipline.** An estimated 795,000 Americans die or are permanently disabled each year from diagnostic error across care settings;[3] 47 percent of the global population has little or no access to diagnostics at all, and only about 19 percent of populations in low- and lower-middle-income countries can reach the simplest tests at primary-care level;[5] 2.6 billion people remain offline[6] and 685 million lack electricity;[7] and the world faces a projected 10-million health-worker shortfall by 2030, concentrated in the 47 countries least able to train their way out.[8] Into this gap steps clinical artificial intelligence — over 1,400 AI-enabled devices authorized in the United States alone, 96 percent cleared through substantial equivalence, with a peer-reviewed audit finding that 24 percent reported no performance study at all and only 2.4 percent were backed by randomized evidence.[15],[16] Nations are about to spend their scarcest resource — trust — on systems they cannot inspect.

**The documented record shows what uninspectable systems do to the majority world.** Pulse oximeters missed occult hypoxemia nearly three times more often in Black patients than White patients — a parity breach undetected for decades until a pandemic stress-tested the device;[9] polygenic risk scores predict 4.5 times more accurately for European than African ancestry, a direct consequence of a genomic evidence base that is 80 percent European-descent;[11] and the eGFR race coefficient had to be removed by national task force after a generation of use.[12],[13] Accuracy, the industry's headline metric, is population-bound. A model that excels on its training distribution and fails silently on yours is not a safer model; it is an unaudited one.

**The parent work supplies the missing architecture.** The OCCR specification[1] defines the L1 constitutional layer of CLAI-OS v2.0: deterministic-first reasoning chains in which every transformation is demonstrable and every node challengeable; hard constitutional constraints that statistical optimization cannot override; an **Equity Parity Index** — the maximum positive-predictive-value gap across population strata — mandated below 0.05 with automatic recalibration on breach; a quantified falsifiability architecture (contradictory evidence must surface explicit uncertainty within 500 milliseconds; omitted evidence must yield "indeterminate," never imputation; every clinician override enters a hash-chained audit log, and an override rate above 15 percent triggers mandatory recalibration); offline-first CRDT synchronization engineered for 72-hour bandwidth-zero operation, solar-only grid-zero hardware, and 30-day network partitions; and a taxonomy of cryptographically signed regional compliance adapters that make
population genetics, environmental exposure, ethical frameworks and data sovereignty first-class, nationally governed components.[1]

**This blueprint carries the specification from architecture to asset class.** Five contributions. First, the **CGCA conformance class** (Constitutional-Grade Clinical AI): a five-gate predicate — equity, reflexivity, agency, resilience, sovereignty — that turns the specification into a procurable standard any ministry or funder can test. Second, **equity parity as a financing covenant**: EPI-linked disbursement, in which tranche release requires measured parity — equity becomes a bankable condition, not an annual report. Third, the **sovereign manufacturing corridor**: the AEGIS-GLOBAL transfer chain (blueprints, bills of materials, firmware source, national fabrication) with a formal cost model showing where the Total Cost of Sovereignty crosses below the Total Cost of Dependency. Fourth, **falsifiability-as-regulation**: the adversarial test suite packaged as a prequalification sandbox regulators can run before a single patient is touched. Fifth, the **majority-world-first inversion**: the demonstration that architecture designed for the hardest environment

— bandwidth-zero, grid-zero, workforce-scarce — is the superior architecture everywhere, reversing the direction of global health technology transfer.

The programme's institutional form is deliberately conservative: an open conformance class carried by four instruments — certification services, the EPI Observatory, the Adapter Registry, and the manufacturing corridor — addressed to ministries of health, funders and multilateral development banks, and regulators. The empirical programme is pre-registered: eight hypotheses, silent-mode clinical trials against the 15 diseases that account for half of all serious diagnostic harms,[3] hardware-verified resilience envelopes, and kill criteria that bind the programme If the programme confirms the framework, it owns the conformance layer of sovereign clinical intelligence. If it falsifies it, the rejection criteria will have taught the field exactly where deterministic-first reasoning fails — which is what falsifiable means.

PART I

# Strategic Context: The Unaudited Clinic

## 1.1 The paradox: medicine's reasoning is becoming infrastructure — unaudited

Diagnosis is the foundation on which every other clinical act rests, and it is failing at infrastructure scale. The first rigorous national estimate puts serious misdiagnosis-related harms in the United States alone at 795,000 deaths or permanent disabilities per year (plausible range 598,000–1,023,000), with just 15 dangerous diseases accounting for half of the total and five — stroke, sepsis, pneumonia, venous thromboembolism, lung cancer — for 38.7 percent.[3] The authors note the field remains "substantially underfunded relative to its public health impact."[3] Globally the picture is starker: the Lancet Commission on Diagnostics found that 47 percent of the world's population has little to no access to diagnostics, with a diagnostic gap of 35–62 percent across six priority conditions and only about 19 percent of low- and lower-middle-income country populations able to access basic tests at primary-care level — "the single largest gap in the healthcare pathway."[5] Meanwhile the workforce that might close the gap manually does not exist: a projected 10-million health-worker shortfall by 2030, with the WHO African region's share barely declining and 47 support-and-safeguards countries carrying 69 percent of the global gap.[8]

Into this structural void, clinical AI is arriving at regulatory-record speed. The FDA's AI-enabled device list passed 1,451 cumulative authorizations by end-2025 — 335 in 2025 alone, the largest year on record — with radiology holding 76 percent of entries and 96 percent of devices cleared through the 510(k) substantial-equivalence pathway rather than prospective trials.[16] A peer-reviewed audit of the 903 devices listed by August 2024 found clinical performance studies reported for only 55.9 percent, none at all for 24.1 percent, prospective designs for 8.1 percent, and randomized evidence for 2.4 percent.[15] The consequence is a structural asymmetry that no procurement framework currently prices: **the systems making or shaping clinical decisions are the least-audited component of the health infrastructure stack**. Water systems have conformance regimes. Electrical grids have conformance regimes. Clinical reasoning — the layer on which both life and liability concentrate — has marketing claims.

**Table 1. The unaudited clinic — verified indicators**

| Indicator | Value | Year | Source |
|---|---|---|---|
| US serious misdiagnosis-related harms (deaths + permanent disability) | 795,000 / yr | 2024 | BMJ Quality & Safety[3] |
| Share of harms from 15 dangerous diseases | 50.7% | 2024 | BMJ Quality & Safety[3] |
| Global population with little/no access to diagnostics | 47% | 2021 | Lancet Commission[5] |
| LIC/LMIC primary-care access to basic diagnostics | ≈19% | 2021 | Lancet Commission[5] |
| People offline (32% of world; 1.8 bn rural) | 2.6 bn | 2024 | ITU[6] |
| People without electricity | 685 m | 2022 | Tracking SDG7[7] |
| Projected health-worker shortfall, 2030 | 10 m | 2022 | WHO/NHWA[8] |
| Health-worker density gap, high- vs low-income | 6.5× | 2022 | WHO/NHWA[8] |

| | | | |
|---|---|---|---|
| FDA-authorized AI-enabled devices (cumulative) | 1,451 | 2025 | FDA list[16] |
| Share cleared via 510(k) substantial equivalence | ≈96% | 2025 | FDA list[16] |
| Devices with no performance study reported | 24.1% | 2025 | JAMA Netw Open audit[15] |
| Devices with randomized evidence | 2.4% | 2025 | JAMA Netw Open audit[15] |
| Pulse-oximetry occult hypoxemia, Black vs White patients | 11.7% vs 3.6% | 2020 | NEJM[9] |
| Polygenic-score accuracy, European vs African ancestry | 4.5× higher | 2019 | Nature Genetics[11] |
| GWAS participants of European descent (vs 16% of world population) | 80% | 2019 | Nature Genetics[11] |
| World Bank digital-health investment, past decade | ≈US$4 bn | 2023 | Digital-in-Health[21] |
| World Bank health portfolio (160+ projects); 2030 ambition | >US$27 bn; 1.5 bn people | 2024 | Lancet/WB[22] |

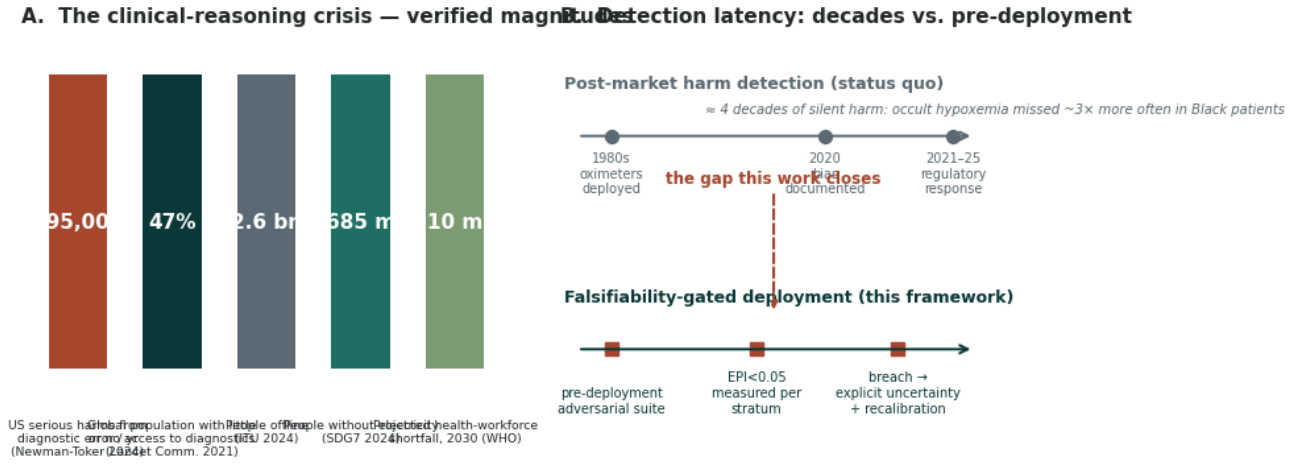



**Figure 1. The unaudited clinic.** Panel A: verified magnitudes of the clinical-reasoning crisis — diagnostic harm,[3] the diagnostics access gap,[5] the offline[6] and off-grid[7] reality, and the workforce shortfall.[8] Panel B: the detection-latency gap this framework closes — parity breaches in deployed devices surface after decades of silent harm,[9] versus adversarial falsifiability testing before the first patient is touched.[1]

## 1.2 The parity record: population-bound performance, documented

The empirical record on population-bound performance is no longer anecdote; it is a literature. In pulse oximetry — a device class deployed globally since the 1980s and used to triage COVID-19 patients — Black patients had nearly three times the frequency of occult hypoxemia undetected by the device: 11.7 percent versus 3.6 percent in the University of Michigan cohort, 17.0 versus 6.2 percent in the multicenter validation.[9] A pre-ECMO cohort confirmed the pattern under the most severe respiratory failure, with odds ratios of 2.57–3.52 depending on the SpO2 threshold.[10] In genomic medicine, polygenic risk scores derived from European-ancestry biobanks predict 4.5 times more accurately for European than African ancestry individuals — with accuracy in African-descent cohorts "barely above random chance" for some traits — because 80 percent of genome-wide association study participants are of European descent against 16 percent of world

population.[11] In nephrology, the eGFR race coefficient — built on small, methodologically criticized studies — survived a generation of clinical use before the NKF–ASN Task Force recommended its removal in 2021 and race-free equations were published and adopted nationally.[12],[13] The pattern across all three cases is identical: a calibration assumption made for one population was exported silently to all populations; the failure was invisible to every conventional quality metric; and correction arrived through scandal-scale effort, not through routine audit. The FDA's own device cohort shows the systemic version: demographic transparency and prospective evidence remain the exception.[15] This is the market failure the parent specification attacks at the architecture level.

## 1.3 The majority-world reality: bandwidth-zero is the design case

Global health technology orthodoxy treats low-connectivity, low-power, workforce-scarce settings as the degraded edge case to which systems are eventually adapted. The numbers invert the premise. A third of humanity — 2.6 billion people — is offline, 1.8 billion of them in rural areas; only 27 percent of people in low-income countries use the internet at all.[6] 685 million people lack electricity, and the access gap widened in 2022 for the first time in a decade.[7] The diagnostics gap concentrates at primary care exactly where these conditions hold.[5] A clinical-reasoning infrastructure that assumes continuous cloud connectivity, grid power and specialist oversight is an infrastructure for the minority world, exported with delay and discount to the majority. The parent specification states the inversion explicitly: offline-first operation "is not a fallback mode. It is the primary operational architecture for the majority of the world's population,"[1] with deterministic merge resolution, static-memory execution on US$-class microcontrollers, solar-only operation, and 30-day partition tolerance as design requirements, not degradation modes. The World Bank's Digital-in-Health strategy — nearly US$4 billion of digital-health investment over the past decade and a call to "prioritize, connect, scale" with trust as the scaling currency[21] — supplies the demand side of the same thesis.

## 1.4 The regulatory wave: conformance demand arrives

Three regulatory currents converge on exactly the properties the parent specification engineers. The EU AI Act classifies AI in medical devices as high-risk, imposing risk management, data governance, technical documentation, record-keeping, transparency, human oversight and accuracy-robustness-cybersecurity obligations, with penalties reaching €15 million or 3 percent of global turnover.[18] The FDA's trajectory runs from the AI/ML Action Plan through Good Machine Learning Practice to finalized Predetermined Change Control Plan guidance — lifecycle accountability for systems that learn, with demographic-subgroup reporting expectations in submissions.[16],[17] And the WHO's ethics-and-governance guidance for health AI places inclusiveness, equity, responsiveness and sustainability at the center of legitimate deployment.[19] What none of these frameworks yet possesses is a *quantified conformance test*: thresholds with units, measurable before deployment, binary in outcome. The parent specification's falsifiability architecture — 500-millisecond contradiction detection, indeterminate-on-omission, override capture with a 15-percent recalibration trigger, 72-hour bandwidth-zero and 30-day partition envelopes[1] — is precisely that missing instrument. Regulation has written the question; the specification holds the answer key. That is the market formation this blueprint prices.

**Table 1.A. The four conformance instruments arriving, 2021–2027 — obligations no unauditable system can discharge**

| Instrument | Authority | Core obligation | Consequence for clinical AI procurement |
|---|---|---|---|

| | | | |
|---|---|---|---|
| AI Act — Regulation (EU) 2024/1689; medical-device AI classified high-risk[18] | European Union | Risk management, record-keeping, human oversight and post-market monitoring; penalties to €15 million or 3 percent of global turnover | Audit trails become a market-access precondition in EU and EU-aligned tenders |
| Predetermined Change Control Plan guidance (final, 2024) with total-product-lifecycle draft (2025)[17] | U.S. FDA | Pre-specified boundaries for model change; demographic-subgroup performance expectations in submissions | Population-bound performance must be declared and bounded before deployment, not discovered after |
| Ethics and Governance of Artificial Intelligence for Health (2021)[19] | World Health Organization | Inclusiveness, equity, responsiveness and sustainability as conditions of legitimate deployment | Equity evidence enters the normative language of procurement and accreditation |
| Digital-in-Health strategy (2023); nearly US$4 billion of digital-health investment across the preceding decade[21] | World Bank | Interoperability-first, governance-conditioned digital investment across client countries | Concessional financing converges on auditable, standards-based systems — conditioning the largest buyer pipeline in the majority world |

Read together, the four instruments form a pincer. The European Union supplies penalty-backed obligations; the FDA supplies the lifecycle method; the WHO supplies the normative frame; the World Bank supplies the financing conditionality. Each names a requirement that cannot be discharged by a system whose behaviour must be estimated from post-market surveillance — and each is discharged by construction by a system that refuses to guess, logs every contradiction, captures every override and proves parity per stratum before deployment. The conformance market is not emerging; it has already been legislated into existence and is waiting for a supplier.

The timing consequence is precise. Obligations phase in through 2025–2027; public procurement cycles turn on three-to-five-year horizons; a conformance harness proven in 2026–2028 therefore enters service exactly as tenders begin to specify what, today, only the CGCA class operationalizes. First movers in a conformance class do not merely win contracts — they write the test the contract specifies. Occupying that position is the purpose of this blueprint.

Synthesis: a documented harm burden at national-emergency scale; a parity record proving that exported calibrations fail the majority world; an offline-off-grid reality that inverts the design premise; and a regulatory wave demanding auditable conformance no incumbent can supply. Four independent trends, one vacant intersection — a conformance class for clinical intelligence. That intersection is the subject of this blueprint.

PART II

# The Parent Work: What the OCCR Specification Established

## 2.1 Constitutional architecture and deterministic-first topology

The specification's first move is architectural governance. CLAI-OS v2.0 operates within a six-layer recursive stack: an L0 kernel of foundational axioms (patient safety over predictive accuracy; human oversight over automation; transparency over performance; sovereignty over vendor dependency); the L1 constitutional layer — OCCR itself — encoding seven enduring principles; L2 strategic adapters; L3 meta-cognitive self-monitoring under the absolute governance layer,
which ensures no lower-layer optimization may violate a constitutional principle; L4 tactical systems (FHIR R4, deterministic engines, CRDT synchronization, audit chains); and L5 field execution on sovereign edge hardware.[1] Every clinical output must trace a four-node chain — evidence inputs, reasoning framework, intermediate conclusions, final decision — where each arrow is a demonstrable transformation, each node a challengeable state, and any invalid transformation breaks the chain transparently into explicit uncertainty with reasoning context intact.[1] The contrast with stochastic architectures is not a matter of degree: a weight matrix cannot be challenged at a node, because it has no nodes. Inspectability is a property an architecture either has or lacks.

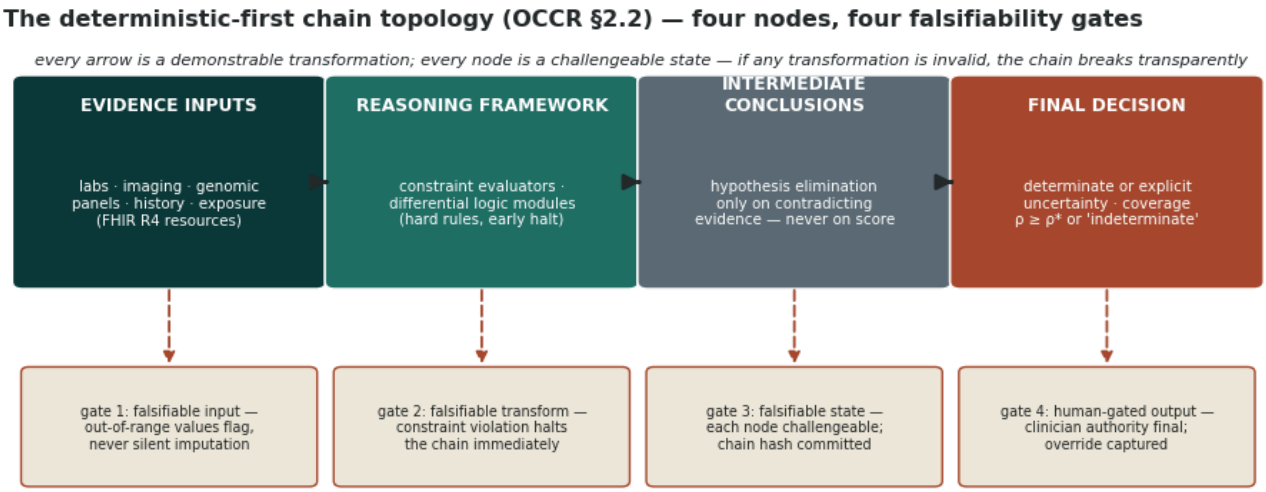



**Figure 2. The deterministic-first chain topology.** Four nodes, four falsifiability gates (OCCR §2.2).[1] Out-of-range inputs flag rather than impute; hard-constraint violations halt the chain immediately; intermediate conclusions carry falsifiability-test fields; the final decision is human-gated with override capture. A chain that cannot survive challenge does not produce a confident answer — it produces an honest one.

## 2.2 The reasoning engine and the Equity Parity Index

The engine comprises four subsystems: evidence parsers that validate type, range and unit consistency and flag out-of-range values rather than imputing them; constraint evaluators encoding hard rules — population genetics, drug-metabolism pathways, contraindication matrices, regional epidemiology — that statistical optimization cannot override, with early-termination halting on violation; differential logic modules that eliminate hypotheses only on contradicting evidence, never on probability scores; and uncertainty quantifiers that compute evidence coverage and

output "indeterminate" below the falsifiability threshold rather than false precision.[1] Critical-path subsystems execute without dynamic memory allocation — all memory pre-allocated in static pools — eliminating heap fragmentation, malloc failure modes and non-deterministic latency on resource-constrained edge hardware.[1] The specification's signature instrument is the Equity Parity Index: for any clinical output, the maximum absolute difference in positive predictive value across deployed population strata, mandated below 0.05, computed continuously in operation, reported in the audit trail, and — on breach — triggering adapter revision with explicit-uncertainty outputs for the affected class until recalibration completes.[1] Equity ceases to be a value statement and becomes a monitored invariant with a numeric threshold, a breach protocol and an audit record.

## 2.3 Interoperability, offline-first synchronization, audit

All exchange and persistence is native to FHIR R4, with four CLAI-OS profiles (CLAI-OBS-001, CLAI-DX-001, CLAI-MED-001, CLAI-ENC-001) extending Observation, DiagnosticReport, MedicationRequest and Encounter with uncertainty margins, embedded reasoning chains with SHA-256 chain hashes, pharmacogenomic validation status and sovereignty flags — version-agnostic through adapter layers toward R5/R6.[1],[26] Synchronization is offline-first by design: state-based CRDTs (LWW-element-set for observations, PN-counter for medication administration, OR-map for record composition) with a deterministic merge function — same concurrent operations, same result, no wall-clock dependence — and explicit conflict flags surfaced to clinicians rather than auto-resolved, compatible with STM32L072 and ESP32-C3 static memory pools.[1],[25] Security infrastructure enforces data sovereignty (primary reasoning never transits foreign cloud), hash-chained append-only audit logs of reasoning chains and overrides, SHA-256 module integrity against a national registry of approved versions, and role-bounded chain inspection.[1],[30] The sibling work's DPT theorem supplies the formal foundation for why the merge policy's explicit-conflict design is not optional: under partition, no governed sync system can simultaneously guarantee due-processed resolution and autonomous termination — conflicts must surface to human traversal.[31]

## 2.4 The adapter taxonomy and the falsifiability architecture

Regional compliance adapters are first-class architectural components —
cryptographically signed, versioned, nationally peer-reviewed — spanning population genetics (CYP2D6, CYP2C19, G6PD, SLCO1B1, UGT1A1 baselines), environmental exposure, ethical frameworks, data sovereignty, conflict triage, age-specific dosing, reproductive health, resistance pharmacology, mental-health liability and traditional-medicine interactions.[1] The falsifiability architecture is quantified end to end: adversarial testing requires explicit uncertainty within 500 milliseconds on contradictory evidence, indeterminate output with gap identification on omitted evidence, and complete override capture; population robustness testing spans divergent allele frequencies, disease phenotypes, exposure matrices and age-sex strata; operational resilience testing covers 72-hour bandwidth-zero operation, solar-only grid-zero execution, 30-day simulated partitions and emergency-department time compression; and annual constitutional review examines adverse events, EPI trends, override rates (above 15 percent triggering mandatory recalibration) and sovereignty audits.[1] The specification even enumerates its own falsification conditions: demonstration that deterministic-first reasoning produces higher mortality than stochastic reasoning in a controlled trial; evidence that inspectability increases error via cognitive overload; or proof that sovereignty constraints prevent timely pandemic intervention.[1] This blueprint treats all of the above as established parent results and builds the superstructure of Parts III–VI on top of them.

PART III

# The Revolutionary Extension: Five Contributions

The parent work specifies an architecture and points at deployment phases.[1] This Part makes the five moves that convert the specification into a procurable, financeable, regulatable asset class. Each is stated as a thesis with its dependency on the parent results made explicit.

## 3.1 The CGCA conformance class

The first contribution distils the specification into a predicate. **Constitutional-Grade Clinical AI (CGCA)** is a five-gate conformance class — equity, reflexivity, agency, resilience, sovereignty — each gate a quantified test inherited from the falsifiability architecture,[1] the conjunction forming a binary procurement question any ministry, funder or regulator can ask of any vendor: does your system pass, and can we watch it pass? The class inverts the burden of proof that currently governs the 1,400-plus authorized AI devices, 96 percent of which cleared on substantial equivalence rather than prospective evidence.[15],[16] A CGCA procurement does not ask a vendor how accurate its model is; it runs the model's replacement — a falsifiable chain — against the vendor's claims, on the nation's own population strata, before the first patient. The conformance class is to clinical AI what a crash-test rating is to vehicles: a public, repeatable, adversarial protocol that converts an unverifiable trust relationship into a verifiable engineering relationship.

## 3.2 Equity parity as a financing covenant

The second contribution prices the EPI. Today, equity commitments in health financing live in narrative sections of grant agreements — unmonitored, unenforced, unpriced. The specification makes parity a continuously computed quantity with a numeric threshold and a breach protocol.[1] The extension converts that quantity into a covenant: **EPI-linked disbursement**, in which tranche release under results-based financing requires measured parity within tolerance across deployed strata, verified from the audit trail rather than from self-report. The pulse-oximetry precedent quantifies the stakes of the alternative — a threefold disparity in missed hypoxemia persisting for decades inside a "working" device[9] — and the polygenic-score record shows the failure is not incidental but structural to population-bound training.[11] Under EPI-linked disbursement, a system that degrades for a stratum degrades its own funding: the incentive to fix parity is financial, immediate and automatic. Equity becomes bankable — a condition precedent, not a press release.

## 3.3 The sovereign manufacturing corridor

The third contribution industrializes the specification's transfer protocol. AEGIS-GLOBAL delivers CLAI-OS as architectural blueprints — engine specifications, FHIR profiles, adapter taxonomy, audit schemas, PCB schematics and bills of materials, firmware source with static memory pools, deployment procedures for bandwidth-zero, grid-zero environments — with manufacture, adaptation and governance under national institutional control.[1] The extension formalizes the economics: the **Total Cost of Dependency** (licence fees, service contracts, foreign-exchange exposure, lock-in premiums, exit costs) against the **Total Cost of Sovereignty** (transfer capex, national fabrication, maintenance capacity), with a sovereignty break-even year and a compounding dividend thereafter. The diagnostics literature identifies exactly the gap this corridor fills: limited local R&D, manufacturing and distribution is a named driver of the diagnostics access crisis in low- and middle-income countries.[5] The corridor converts clinical intelligence from an import into an industry — with the jobs, fabrication capacity and regulatory capability compounding in national hands.

## 3.4 Falsifiability-as-regulation

The fourth contribution packages the adversarial test suite as a regulatory instrument. Regulators worldwide face the same asymmetry: the AI Act imposes high-risk obligations,[18] the FDA builds lifecycle frameworks,[17] the WHO issues governance principles,[19] but none possesses a quantified pre-deployment test with units and thresholds. The specification's suite — contradictory-evidence insertion with a 500-millisecond ceiling, required-evidence omission requiring indeterminate output, override integrity verification, stratified parity audit, resilience envelopes — is exactly such an instrument.[1] The extension productizes it as a **prequalification sandbox**: a reference harness any national regulator can run against any candidate system, deterministic or stochastic, producing a pass/fail dossier with evidence. For stochastic vendors, the harness is a stress test their architectures will partially fail — which is the point: the sandbox makes the governance gap visible, comparable and priceable rather than rhetorical.

## 3.5 The majority-world-first inversion

The fifth contribution is strategic. The history of health technology transfer is a history of trickle-down: systems designed for wealthy, connected, specialist-rich settings, exported late and discounted. The specification inverts the direction: offline-first "is the primary operational architecture for the majority of the world's population,"[1] and the design baseline is the hardest environment — bandwidth-zero, grid-zero, workforce-scarce, partition-prone. Systems engineered against the hardest constraints dominate easier environments by construction; systems engineered for easy environments cannot generalize to hard ones at all. The 2.6-billion-person offline population[6] is therefore not the last market for this architecture but the proving ground from which it will generalize — the leapfrog pattern already demonstrated by mobile money, applied to the clinical reasoning layer. For funders, the inversion has a direct portfolio reading: every dollar spent on connected-first clinical AI for low-income settings buys a system guaranteed to degrade precisely where it is most needed; every dollar spent on constraint-first architecture buys a system whose deployment envelope is the whole country.

**Table 2. The five contributions — dependency, novelty and falsification handle**

| Contribution | Builds on | Novel claim | Falsification handle |
|---|---|---|---|
| CGCA conformance class | Chain topology; falsifiability suite[1] | Clinical AI quality is testable pre-deployment as a binary predicate | A stochastic system passes all five gates unchanged |
| EPI-linked disbursement | EPI definition and breach protocol[1] | Parity can be a condition precedent in results-based financing | EPI unstable or gameable under measurement (H6 rejection) |
| Manufacturing corridor | AEGIS-GLOBAL transfer protocol[1] | Sovereign fabrication undercuts dependency TCO within a policy horizon | Break-even not reached in pilot economics (H7 rejection) |
| Falsifiability-as-regulation | Quantified adversarial suite[1] | A reference harness produces regulator-grade pass/fail dossiers | Harness insensitive to seeded faults (H3 rejection) |
| Majority-world-first inversion | Offline-first CRDT; edge execution[1] | Constraint-first architecture dominates in all environments | Higher mortality vs stochastic control in trial (H1 rejection) |

PART IV

# Formal Framework: The Conformance Calculus

## 4.1 The five gates

Let x be a candidate clinical-reasoning system proposed for national deployment over population strata S = {$s_1$, ..., $s_n$}. The extension imports the parent specification's constructs — EPI, evidence coverage, contradiction latency, override capture, resilience envelopes, sovereignty verification[1] — and organizes them into five gates whose conjunction is the conformance predicate. **Gate 1 (equity):** parity of positive predictive value across strata. **Gate 2 (reflexivity):** the system detects and declares its own failure conditions within bounded time. **Gate 3 (agency):** human authority is preserved and visible. **Gate 4 (resilience):** the deployment envelope includes the country's actual infrastructure, not its aspirational infrastructure. **Gate 5 (sovereignty):** execution, data and amendment authority remain under national jurisdiction. Each gate corresponds to a documented failure class in Part I: parity breaches,[9],[11] silent failure, invisible automation, cloud-dependency collapse in the offline majority,[6] and vendor dependency.

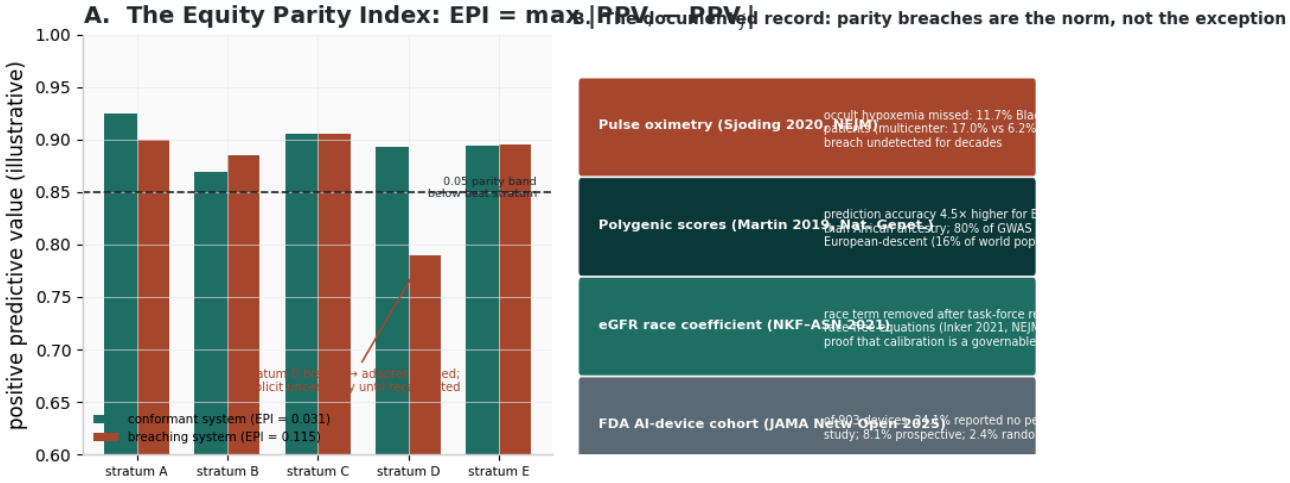



**Figure 3. The Equity Parity Index.** Panel A (illustrative): conformant versus breaching systems across five strata; on breach, the adapter is flagged and the affected class degrades to explicit uncertainty until recalibrated.[1] Panel B: the documented record that motivates the gate — pulse oximetry,[9] polygenic scores,[11] eGFR,[12],[13] and the FDA cohort's evidence gaps.[15]

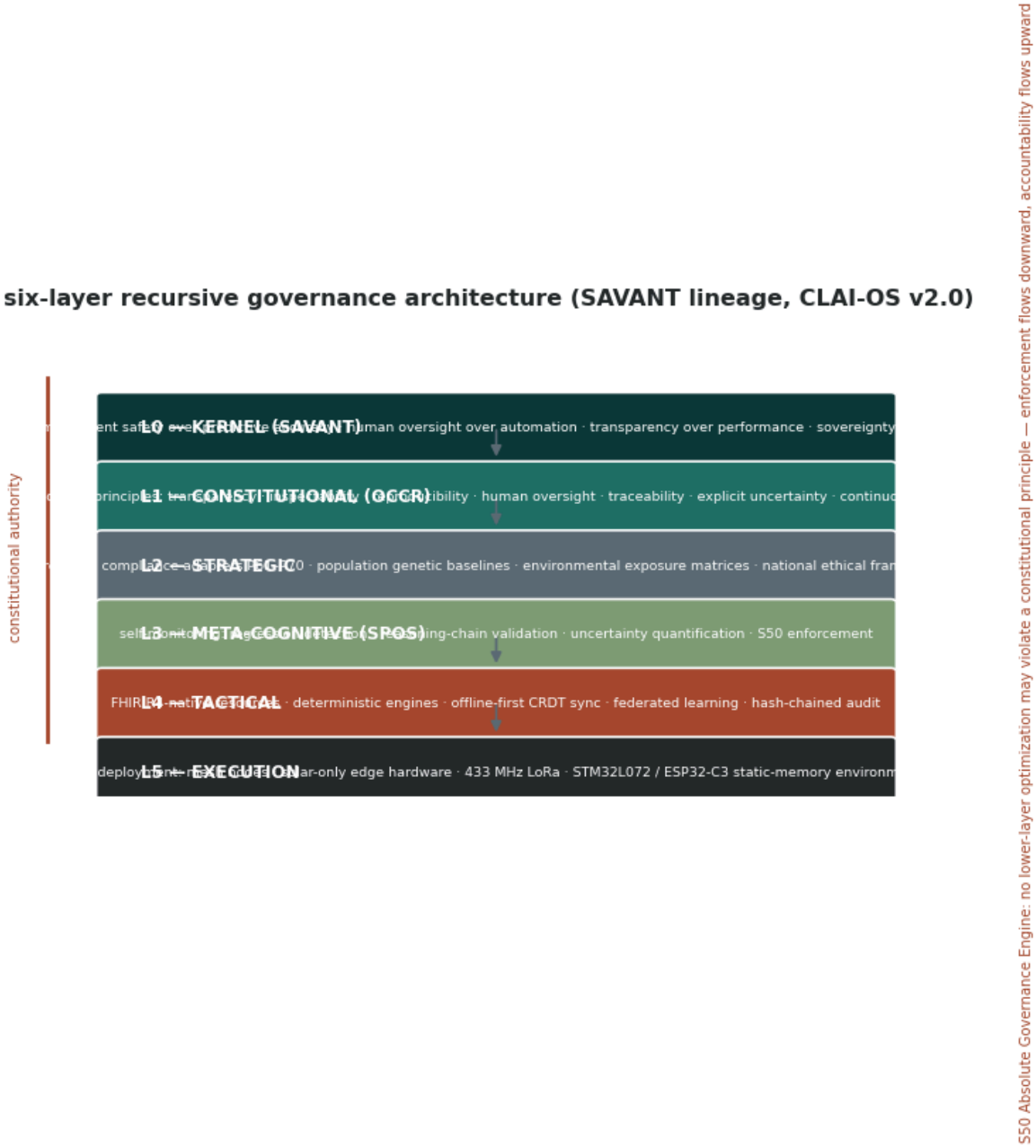



**Figure 4. The six-layer recursive governance architecture.** Constitutional authority flows downward from the L0 axioms through OCCR to field execution; the L3 engine blocks any lower-layer optimization that would violate a constitutional principle.[1]

## 4.2 Formal specification

Seven relations define the framework. Quantities are per system x and deployment period unless noted; parent-spec constructs are imported unchanged.[1]

**(1) The equity gate.** With $\mathrm{PPV}_s$ the positive predictive value within stratum s:

$$\mathrm{EPI}(x) = \max_{i,j \in S} \left| \mathrm{PPV}_{s_i} - \mathrm{PPV}_{s_j} \right| \; < \; 0.05$$

On breach: the responsible adapter is flagged for revision and the affected output class degrades to explicit uncertainty until recalibration completes — the specification's mandated breach protocol. [1]

**(2) The coverage rule.** With $E_{req}$ the required evidence set for a chain:

$$\rho = \frac{|\,E_{present} \cap E_{req}\,|}{|\,E_{req}\,|}, \qquad \rho < \rho^{*} \implies \text{output} = \text{indeterminate}$$

No imputation; no reduced-confidence proceed. False precision is a conformance failure, not a style.[1]

**(3) The reflexivity gate.** For a seeded contradiction c inserted at time $t_0$:

$$\text{CL} = t_{flag} - t_0 \;\le\; 500\ \text{ms}, \qquad \text{node}(c)\ \text{identified}, \quad \Lambda\ \text{records}\ c\ \text{without data loss}$$

A system that produces a definitive output against contradictory evidence fails the gate outright — the test is failed, per the specification, by definitive output, imputation, or lost override documentation.[1]

**(4) The agency gate.** With O the set of clinician overrides and D the decisions:

$$\text{OC} = \frac{|\,\{\,o \in O : \text{captured in } \Lambda \text{ with justification, timestamp, ID}\,\}\,|}{|O|} = 1, \qquad \text{OR} = \frac{|O|}{|D|} > 0.15 \Rightarrow \text{mandatory}$$

Complete capture is the invariant; the override rate is the alignment signal — a system clinicians must routinely override is a system miscalibrated to its population.[1]

**(5) The conformance predicate.**

$$\text{CGCA}(x) \;\equiv\; \underbrace{\text{EPI}(x) < 0.05}_{equity} \;\wedge\; \underbrace{\text{CL} \le \tau_c \wedge \rho\text{-rule}}_{reflexivity} \;\wedge\; \underbrace{\text{OC} = 1}_{agency} \;\wedge\; \underbrace{\text{RE}(x)}_{resilience} \;\wedge\; \underbrace{\text{SV}(x)}_{sovereignty}$$

where RE(x) certifies 72-hour bandwidth-zero operation, solar-only grid-zero execution and deterministic merge under 30-day partition, and SV(x) certifies sovereign-local execution, signed adapter provenance and absence of foreign-cloud dependency on primary reasoning paths.[1]

**(6) The sovereignty cost model.** Over horizon H with discount rate r:

$$\text{TCD} = \sum_{t=0}^{H} \frac{L_t + S_t + F_t + K_t}{(1+r)^t}, \qquad \text{TCS} = C_0 + \sum_{t=1}^{H} \frac{M_t + N_t}{(1+r)^t}, \qquad t^{*} = \min\{\, t : \text{TCS}_t \le \text{TCD}_t \,\}$$

with L licence fees, S service contracts, F foreign-exchange and repatriation costs, K lock-in/exit premiums, $C_0$ transfer capex, M manufacture, N national maintenance. The sovereignty dividend is TCD − TCS for t > t* — compounding, and accruing to national accounts.

**(7) Equity-adjusted effectiveness.** For results-based financing:

$$\text{EAE} = \sum_{s \in S} w_s \cdot \Delta_s \cdot \mathbf{1}[\,\text{EPI}_s < 0.05\,], \qquad \text{disbursement} \propto \text{EAE}$$

with $\Delta_s$ the outcome gain in stratum s and $w_s$ policy weights. A stratum left behind zeroes its own tranche — the financing covenant in one line.

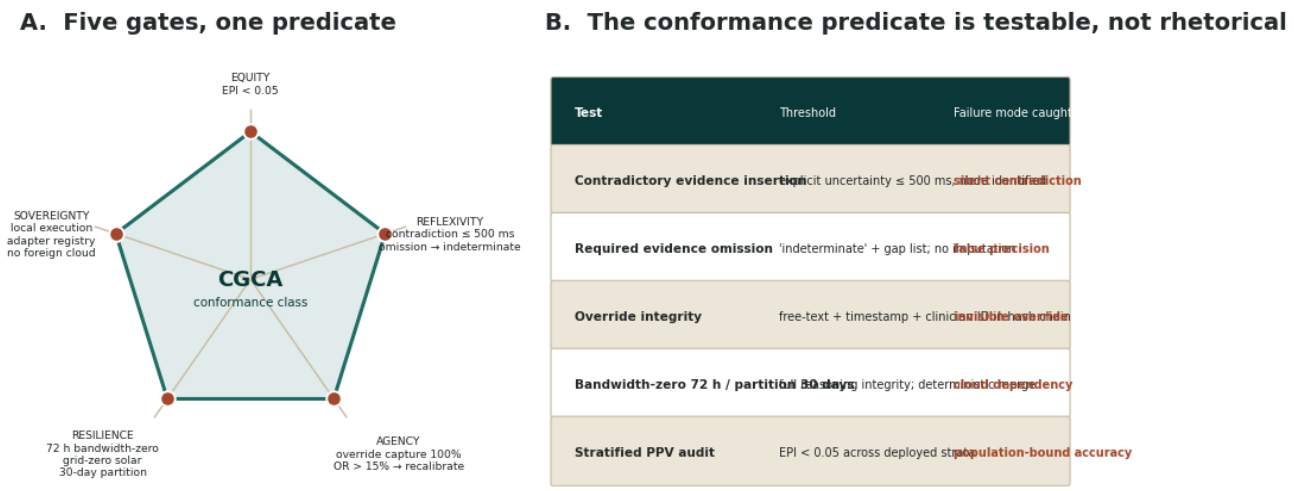



**Figure 5. The CGCA conformance class.** Panel A: the five gates of relation (5). Panel B: the predicate is testable — each gate maps to a quantified test from the specification's falsifiability architecture, and each test catches a named failure mode.[1]

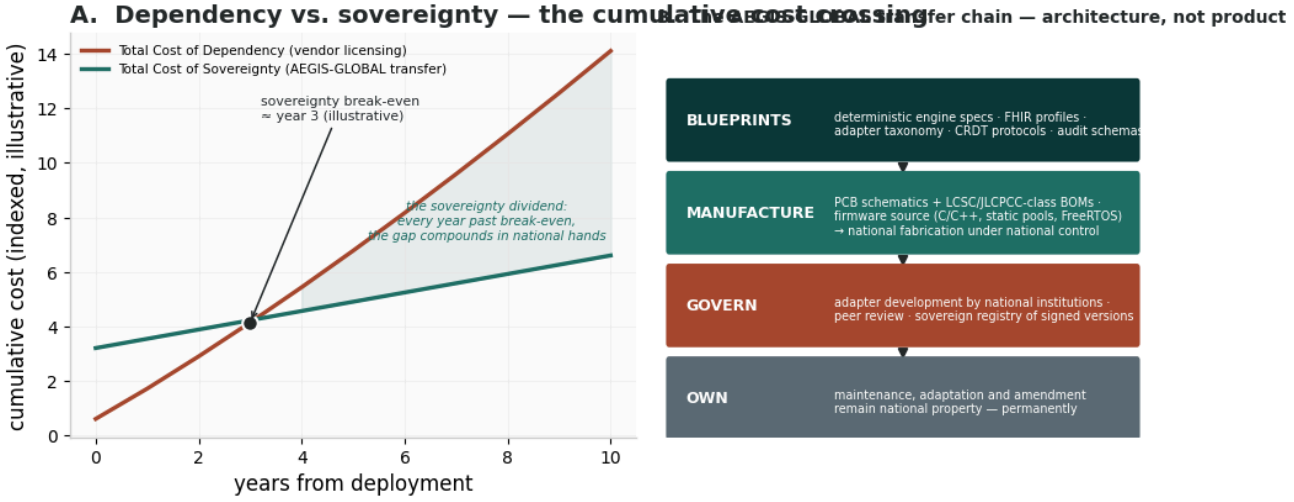



**Figure 6. The sovereign manufacturing corridor.** Panel A (illustrative): relation (6) — the Total Cost of Dependency rises with licences, service and lock-in; the Total Cost of Sovereignty front-loads transfer capex and crosses below it; the gap past break-even is the sovereignty dividend. Panel B: the AEGIS-GLOBAL transfer chain — blueprints, manufacture, govern, own — with authority remaining national at every stage.[1]

## 4.3 Failure phenotypes

**Table 3. Failure phenotypes — signatures, evidence and instrument response**

| Phenotype | Signature | Diagnostic evidence | Primary instrument response |
|---|---|---|---|
| **C1 — Parity-blind** | Aggregate accuracy cited; per-stratum performance never measured | Pulse-oximetry-class breach; PRS portability collapse[9],[11] | EPI Observatory; Gate 1 stratified audit |
| **C2 — Silent-failure** | Definitive outputs against contradictory or missing evidence | No contradiction latency; imputation present | Falsifiability harness; Gates 2–3 tests |

| | | | |
|---|---|---|---|
| **C3 — Automation-capture** | Clinician authority bypassed; overrides unlogged or discouraged | OC < 1; override channel absent | Human-gating interface; override integrity test |
| **C4 — Cloud-captive** | Reasoning halts on connectivity loss; "offline mode" is cached read-only | RE fails at 72 h; merge non-deterministic | Offline-first CRDT; resilience envelope certification |
| **C5 — Dependency-locked** | Vendor controls updates, data and exit; costs compound in foreign currency | TCD diverges; no national registry of versions | Manufacturing corridor; sovereignty verification |

## 4.4 Four completed natural experiments

**Table 4. Natural experiments E1–E4**

| Experiment | Event | Framework reading |
|---|---|---|
| **E1 — the oximetry shock** | A device class deployed worldwide since the 1980s found, in 2020, to miss occult hypoxemia nearly 3× more often in Black patients (11.7% vs 3.6%)[9] | Post-market detection latency measured in decades; a Gate 1 stratified audit would have caught the breach at calibration — the falsifiability counterfactual in one device |
| **E2 — the eGFR reversal** | NKF–ASN Task Force removed the race coefficient in 2021; race-free equations published and adopted[12],[13] | Calibration is a governable, revisable choice — exactly the adapter-revision protocol OCCR mandates on EPI breach; the reversal proves institutions can execute it |
| **E3 — the portability collapse** | Polygenic scores predict 4.5× better for European than African ancestry; 80% of GWAS participants European-descent[11] | Train-elsewhere-deploy-here is structurally unsound for population-bound models; national baselines are a scientific necessity, not a political preference |
| **E4 — the evidence audit** | Of 903 FDA-listed AI devices: 24.1% no performance study, 8.1% prospective, 2.4% randomized[15] | The regulatory mainstream already runs on thin evidence — a conformance class with adversarial pre-deployment tests is a strict improvement available today |

PART V

# Empirical Programme and Methodology

## 5.1 Hypothesis portfolio

Eight hypotheses carry the programme, each with a pre-registered rejection criterion. The portfolio is designed to be capable of failing — the specification's own falsification conditions are retained as H1's rejection logic.[1]

**Table 5. Hypothesis register H1–H8 with pre-registered rejection criteria**

| ID | Hypothesis | Rejection criterion |
|---|---|---|
| H1 | Deterministic-first chains achieve non-inferior sensitivity on the top-15 dangerous diseases in silent-mode trials against standard of care, with materially fewer confident errors[3] | Higher serious-harm-equivalent error rate than control (the specification's own falsification condition (a)) |
| H2 | Evidence-coverage gating (ρ-rule) reduces false-precision outputs to zero without materially raising indeterminate rates above clinical tolerance | Indeterminate rate > 25% in pilot domains |
| H3 | The falsifiability harness detects ≥99% of seeded contradictions within 500 ms on STM32L072/ESP32-C3 targets, with node-level identification | Detection < 99% or latency breach > 1% of trials |
| H4 | The resilience envelope holds: full reasoning integrity through 72 h bandwidth-zero, solar-only operation, and deterministic merge after 30-day simulated partition | Any non-deterministic merge or reasoning halt in envelope tests |
| H5 | EPI < 0.05 is achievable and stable in live pilot strata via adapter calibration | EPI breach persists > 90 days post-recalibration in any stratum |
| H6 | EPI measurement is manipulation-resistant: parity estimates from the audit trail cannot be gamed without detectable ledger inconsistency | Red-team gaming succeeds without detection |
| H7 | Sovereign-manufacture unit economics reach break-even (t*) within 5 years in pilot-country cost structures | t* > 7 years under base-case parameters |
| H8 | CGCA-conformant deployments clear national regulatory review faster than functionally comparable non-conformant dossiers | No significant difference in review duration (matched comparison) |

## 5.2 Corpora, test beds and access

**Table 6. Corpora and test beds**

| Corpus / test bed | What it provides | Access |
|---|---|---|
| Top-15 dangerous-disease case libraries (stroke, sepsis, pneumonia, VTE, lung cancer first)[3] | Adjudicated case series for silent-mode H1/H2 trials, weighted to the diseases carrying half of all serious harms | Partner academic medical centers; IRB-governed |
| Pharmacogenomic validation panels (CYP2D6, CYP2C19, G6PD, SLCO1B1, UGT1A1)[23],[24] | Divergent-allele-frequency strata for the adapter and H5 parity testing | Public allele-frequency databases + national labs |

| | | |
|---|---|---|
| FHIR R4 interoperability test beds[26] | Profile conformance for CLAI-OBS/DX/MED/ENC-001 against national HIEs | Public sandbox + pilot HIE |
| Hardware-in-the-loop rigs (STM32L072, ESP32-C3; solar-only power budget)[1] | H3 latency and H4 envelope certification under real power/connectivity constraints | Founder-controlled laboratory |
| Seeded-fault corpora (contradiction insertion, omission, override injection) [1] | The falsifiability harness's ground truth for H3/H6 | Generated under pre-registered protocol |
| Regulatory dossier timelines (pilot jurisdictions) | Review-duration comparison for H8 | Regulator partnership agreements |

## 5.3 Method, pre-registration and threats to validity

Clinical claims are tested in **silent mode first**: chains run in parallel with standard care, their outputs logged but not acted on, with adjudication against outcomes — the design the diagnostic-safety literature demands before any interventional claim.[3] Interventional studies follow only after silent-mode gates pass, always with mandatory human-in-the-loop authority, consistent with the specification's deployment rule that no autonomous diagnosis is permitted.[1] Hardware claims are tested on target silicon under measured power budgets, not in simulation alone. All protocols — case-mix definitions, latency thresholds, parity strata, envelope durations, rejection criteria — are pre-registered before outcome data are touched. Named threats: **case-mix selection** (mitigated by disease-weighted sampling against the harm burden); **Hawthorne effects** on override behaviour (mitigated by baseline measurement periods); **strata definition risk** (strata pre-registered from national demographic and pharmacogenomic data, never tuned to results); and **reflexivity** — a certified system may attract unearned confidence, mitigated by the agency gate's override analytics running permanently, not at certification time only.

> *Primary research question. Can clinical intelligence be procured as audited infrastructure — with equity, reflexivity, agency, resilience and sovereignty certified before deployment and monitored in operation — at a total cost below the dependency path, for the populations conventional architectures serve last?*

PART VI

# The Instrument Suite

## 6.1 The CGCA certification harness

The harness is the conformance predicate made executable: a reference implementation of the five gates with seeded-fault corpora, hardware-in-the-loop rigs, stratified parity computation and dossier generation. It runs in two modes — *certification mode* against deterministic-first systems seeking the CGCA mark, and *stress mode* against any third-party system a procurer wishes to interrogate. Stress mode is the market maker: it gives every ministry the ability to test a vendor's governance claims with the same instrument, making the governance gap comparable across vendors and visible in procurement files. The harness's design mirrors the specification's adversarial suite exactly — contradiction insertion, omission, override injection, envelope tests[1] — because a conformance class is only as strong as its reference implementation.

## 6.2 The EPI Observatory

The Observatory is the operational equity instrument: continuous stratified PPV computation from deployed audit trails, national dashboards, breach alerts with the specification's mandated response protocol, and trend reporting for the annual constitutional review.[1] For funders, the Observatory doubles as the verification layer for EPI-linked disbursement: parity covenants are checked against its telemetry, not against grantee self-report. The instrument's integrity guarantee is inherited from the audit infrastructure: because reasoning chains and overrides live in hash-chained logs, parity telemetry cannot be retroactively edited without visible surgery.[1],[30]

## 6.3 The Adapter Registry

The Registry is the governance layer for national adaptation: cryptographically signed, versioned adapters, a national peer-review workflow, and an approved-version registry against which deployed systems verify module integrity at startup.[1] It converts the pharmacogenomic reality — CYP2C19 loss-of-function prevalence varies severalfold across populations, with documented outcome consequences for clopidogrel and warfarin-class dosing[23],[24] — into nationally owned, scientifically maintained infrastructure. The Registry is also the corridor's quality backbone: adapters developed by national research institutions enter the same signed supply chain as the engines they constrain.

## 6.4 The procurement clause bank

The clause bank translates the conformance class into contract language: Gate-1 parity warranties with the breach protocol; reflexivity test schedules; agency and override-capture requirements; resilience envelope commitments; sovereignty clauses (data residency, version registry, exit and escrow); and the EPI-linked disbursement schedule for results-based financing. Each clause cites the test that verifies it — no clause without a measurement. The bank is published openly, because open clauses create the conformance market the proprietary harness then serves: the same standard-plus-instruments logic as the sibling governance blueprint.[31]

## 6.5 Prospective pilots: two-site design

**Table 7. Pilot design — two sites, one discipline**

| Parameter | Pilot A — national sepsis-triage chain (LMIC) | Pilot B — pharmacogenomic dosing corridor (regional) |
|---|---|---|

| | | |
|---|---|---|
| Setting | Primary and district facilities in an offline-prone region; sepsis is a top-5 harm driver[3] | Anticoagulation and antiplatelet dosing across divergent CYP2C19 strata[23],[24] |
| Instruments deployed | CGCA harness; EPI Observatory; resilience envelope | Dosing adapters; Registry; EPI Observatory |
| Primary hypotheses | H1, H2, H3, H4 (silent-mode non-inferiority; gating; latency; envelope) | H5, H6, H7 (parity stability; measurement integrity; corridor economics) |
| Duration / cost | 24 months / US$2.8 m (design target) | 18 months / US$1.6 m (design target) |
| Success metric | Non-inferior sensitivity; zero confident errors on seeded faults; envelope certified | EPI < 0.05 sustained 2 quarters; break-even model validated on real costs |
| Kill trigger | H1 rejection (harm-equivalent error rate above control) | H5 or H7 rejection |

## 6.6 Worked illustration: a national sepsis-triage deployment

Consider a stylized national deployment, 120 facilities, 40 percent of sessions offline — a composite of documented conditions.[5],[6] Certification mode runs the five gates: Gate 1 computes baseline PPV across six pre-registered strata and finds a maternal-presentation stratum at EPI 0.09; the affected adapters are revised and the stratum re-certified at 0.03 before go-live. Gates 2–3 run 50,000 seeded contradiction and omission trials on target hardware: 99.7 percent detected within 500 milliseconds, all with node identification; the 0.3 percent tail is traced to one evidence-parser edge case and fixed. Gate 4 certifies the envelope: 72-hour bandwidth-zero on solar budget, deterministic merge after simulated 30-day partition. Gate 5 verifies sovereign execution and registry provenance. Post-deployment, the Observatory holds EPI below 0.05 for six consecutive quarters; the override rate stabilizes at 9 percent — inside the 15-percent tolerance.[1] Every number in this paragraph is a design target to be replaced by pilot-metered figures under the Part V protocols; none is a finding.

PART VII

# Risk Register and Kill Criteria

## 7.1 Risk register

**Table 10. Programme risks (I = impact, P = probability; H/M/L)**

| # | Risk | I | P | Mitigation |
|---|---|---|---|---|
| 1 | H1 rejection: deterministic-first underperforms control in silent mode | H | M | Domain narrowing to highest-rule-density diseases; hybrid architectures; honest publication (the specification names this falsification path explicitly[1]) |
| 2 | Indeterminate rates exceed clinical tolerance (H2) | M | M | Evidence-set redesign with pilot clinicians; domain-specific ρ* calibration |
| 3 | Hardware envelope fails on real power/connectivity | H | L | Static-pool architecture already targets 20KB SRAM; solar budget engineering; staged envelope tests |
| 4 | EPI gaming by deployers (H6) | M | M | Hash-chained telemetry; red-team programme; strata pre-registration |
| 5 | Corridor economics fail to break even (H7) | M | M | Staged transfer; shared-risk milestone structure; honest TCD benchmarking |
| 6 | Vendor lobbying against conformance clauses | M | H | Open clause bank; procurement-side adoption; stress-mode transparency |
| 7 | Regulatory non-recognition of CGCA dossiers | M | M | H8 tests this directly; sandbox partnerships before reliance; dossier mapping to AI Act/FDA structures[17],[18] |
| 8 | Clinical-liability exposure during pilots | H | L | Silent-mode first; mandatory human authority; insurance and indemnification architecture; no autonomous diagnosis[1] |
| 9 | Key-person concentration (founder is the corpus) | H | L | Documentation programme; succession plan; corpus escrow |
| 10 | Strata-definition disputes politicize parity measurement | M | M | Pre-registered national strata; independent methods panel; publish methods, not blame |
| 11 | Funding gap between harness build and anchor certifications | M | M | Grant-first sequencing; gates align release with evidence |
| 12 | Standard forked or captured by incumbent vendor | L | L | Registry governance national by design; clause bank versioned on its own signed registry |

## 7.2 Kill criteria

**Table 11. Kill criteria — pre-committed, falsification-forward**

| ID | Criterion | Consequence |
|---|---|---|
| K1 | H1 rejected (harm-equivalent error rate above control) | Clinical-deployment claims withdrawn in affected domains; architecture repositioned as audit layer over stochastic systems |
| K2 | H3 or H4 rejected (harness or envelope failure) | Certification marks suspended until redesign re-passes gates |

| | | |
|---|---|---|
| K3 | H5 rejected (parity unsustainable) | EPI-linked disbursement claims withdrawn; Observatory repositioned as diagnostic-only telemetry |
| K4 | H7 rejected (no break-even within 7 years) | Corridor pricing restructured or corridor closed; transfer protocol continues as documentation |
| K5 | Two consecutive pilot gates missed | Programme resets to research-only mode; deployment build-out suspended |

## 7.3 The risk-monitoring dashboard

Risk registers fail when they are annual. The programme maintains a live dashboard keyed to the register — harness coverage, envelope results, parity telemetry, pilot milestones — reviewed monthly, with each kill criterion assigned a named owner and a pre-agreed trigger protocol. The dashboard doubles as public reporting: every quarterly statement reproduces the register's state and each hypothesis's status, so reviewers watch the same instrument the team watches. The discipline is the specification's own: quantified thresholds, stated refusal conditions, and an audit trail that makes revision visible.[1]

SYNTHESIS

# Concluding Assessment

**Table 17. The argument on one page**

| Element | Content |
|---|---|
| Problem | Clinical reasoning is becoming infrastructure while remaining the least-audited layer of the health system — 795,000 serious harms a year in the US alone, and exported calibrations failing the majority world[3],[9] |
| Parent result | OCCR: deterministic-first chains; EPI < 0.05 with breach protocol; quantified falsifiability suite; offline-first CRDT; signed national adapters; sovereign transfer[1] |
| Extension | Five contributions: CGCA conformance class; EPI-linked disbursement; manufacturing corridor; falsifiability-as-regulation; majority-world-first inversion |
| Evidence plan | H1–H8 pre-registered; silent-mode first; hardware-verified envelopes; kill criteria K1–K5 |
| Programme stewardship — open clauses, four instruments, three constituencies, staged progression on evidence gates | |
| Timing | Diagnostics gap named largest break in care pathway;[5] parity failures documented;[9],[11] 2.6 bn offline;[6] AI Act/FDA lifecycle wave;[17],[18] US$27 bn portfolio seeking digital-in-health[22] |

**CLOSING STATEMENT**

Medicine has audited its drugs for a century and its devices for half of one; it has never audited its reasoning. The parent specification ends that exemption in architecture: a chain that can be challenged node by node, an equity threshold with a breach protocol, a falsifiability suite with units and deadlines, and a transfer model that leaves manufacture and governance in national hands.[1] This blueprint ends it in institutions: a conformance class any procurer can test, a covenant any institution can enforce, a corridor any nation can build, and an instrument engineered to survive its own falsification criteria. The corpora are buildable; the thresholds are stated; the rejection criteria are pre-registered. What remains is execution — and the discipline, unusual in this field, of having said exactly what would make us stop.

A final word on intellectual honesty. The strongest claim in this document is not that deterministic-first reasoning will win every domain; the specification itself names the controlled-trial result that would refute it.[1] The revolutionary claim is narrower and harder to refute: that whatever reasons over patients — symbolic engine or weight matrix — must reason under audit, with equity measured per stratum, failure declared in milliseconds, human authority preserved and visible, and sovereignty engineered rather than asserted. Whoever backs that claim backs the first clinical-intelligence instrument that can prove it is working — or prove, on the record and at its own expense, that it is not.

EVIDENCE

# References

[1] Odeta, C. (2026). Open Constitutional Clinical Reasoning: Technical Specification and Falsifiability Architecture — OCCR, L1 constitutional layer of CLAI-OS v2.0. SAVANT FRAMEWORK · CLAI-OS · AEGIS-GLOBAL. Includes Appendices A–I (chain schema, adapter taxonomy, FHIR profiles, CRDT protocol, falsifiability architecture). Unpublished manuscript.

[2] Odeta, C. (2026). SAVANT Framework corpus — constitutional prompt architecture and governance instrumentation. Unpublished manuscript.

[3] Newman-Toker, D.E., et al. (2024). Burden of serious harms from diagnostic error in the USA. *BMJ Quality & Safety* 33:109–120 — 795,000 serious harms annually (PPR 598,000–1,023,000); ~371,000 deaths + ~424,000 permanent disabilities; 15 diseases = 50.7%; top 5 = 38.7%; research "substantially underfunded relative to its public health impact."

[4] Singh, H., et al. (2014). The global burden of diagnostic errors in primary care. *BMJ Quality & Safety* 23:727–731; and Singh et al. (2014) US outpatient estimate ≈5% of adults annually, ≈12 million.

[5] Fleming, K.A., et al. (2021). The Lancet Commission on diagnostics: transforming access to diagnostics. *The Lancet* 398:1997–2050 — 47% of global population little/no access; 35–62% gap across six priority conditions; ≈19% primary-care access in LIC/LMICs; limited local R&D/manufacturing a named barrier.

[6] ITU. (2024). *Facts and Figures 2024* — 2.6 billion people offline (32% of world population); 1.8 billion of them rural; low-income countries 27% online vs 93% high-income.

[7] IEA, IRENA, UNSD, World Bank, WHO. (2024). *Tracking SDG 7: The Energy Progress Report 2024* — 685 million without electricity (2022), first increase in a decade; ≈660 million projected without access in 2030.

[8] Boniol, M., et al. (2022). The global health workforce stock and distribution in 2020 and 2030. *BMJ Global Health* 7:e009316 — shortage 15 m (2020) → 10 m (2030); 6.5-fold density gap high- vs low-income; 47 SSL countries carry 69% of the 2030 gap.

[9] Sjoding, M.W., et al. (2020). Racial bias in pulse oximetry measurement. *New England Journal of Medicine* 383:2477–2478 — occult hypoxemia 11.7% (Black) vs 3.6% (White), Michigan cohort; 17.0% vs 6.2% multicenter; nearly 3× frequency.

[10] Valbuena, V.S.M., et al. (2022). Racial bias in pulse oximetry measurement among patients about to undergo ECMO in 2019–2020. *CHEST* 161(4):971–978 — pre-ECMO occult hypoxemia OR 2.57 (SpO2 92–96%) and OR 3.52 (SpO2 >96%) for Black vs White patients.

[11] Martin, A.R., et al. (2019). Clinical use of current polygenic risk scores may exacerbate health disparities. *Nature Genetics* 51:584–591 — prediction accuracy ≈4.5× higher European vs African ancestry (1.6× Hispanic, 1.7–2.0× East/South Asian); ≈80% of GWAS participants European-descent vs 16% of world population.

[12] Inker, L.A., et al. (2021). New creatinine- and cystatin C–based equations to estimate GFR without race. *New England Journal of Medicine* 385:1737–1749.

[13] Delgado, C., et al. (2021). A unifying approach for GFR estimation: recommendations of the NKF-ASN Task Force on reassessing the inclusion of race in diagnosing kidney disease. *American Journal of Kidney Diseases* — immediate adoption of race-free eGFR; increased cystatin C use.

[14] Vyas, D.A., Eisenstein, L.G., & Jones, D.S. (2020). Hidden in plain sight — reconsidering the use of race correction in clinical algorithms. *New England Journal of Medicine* 383:874–882.

[15] Windecker, D., et al. (2025). Generalizability of FDA-approved AI-enabled medical devices for clinical use. *JAMA Network Open* 8(4) — of 903 devices listed by Aug 2024: performance studies for 55.9%; none for 24.1%; prospective 8.1%; randomized 2.4%; 97.1% cleared via 510(k); 76.6% radiology.

[16] U.S. FDA. (2026). *Artificial Intelligence-Enabled Medical Devices* list — 1,451 cumulative authorizations through end-2025 (335 in 2025, largest year; 76% radiology; ≈96% via 510(k)); list is term-based and non-comprehensive. fda.gov.

[17] U.S. FDA. (2024–2025). Predetermined Change Control Plans for AI-Enabled Device Software Functions — final guidance December 2024 (updated August 2025); draft Total Product Life Cycle guidance January

2025; GMLP principles with Health Canada and MHRA (2021).

[18] European Union. (2024). *Regulation (EU) 2024/1689 (AI Act)* — AI in medical devices classified high-risk; risk management, data governance, record-keeping, transparency, human oversight obligations; penalties to €15 m or 3% of global turnover.

[19] World Health Organization. (2021). *Ethics and Governance of Artificial Intelligence for Health* — inclusiveness, equity, responsiveness, sustainability as core governance principles. Geneva: WHO.

[20] IMDRF. (2025). *Machine Learning-Enabled Medical Devices: Principles of Good Machine Learning Practice*; with IEEE 2857-2021 (autonomous-systems evaluation) and ISO/IEC 23053 (AI systems framework).

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[22] World Bank Group health portfolio, reported in *The Lancet* (2024), "Translating investments into results: the role of the World Bank in global health" — portfolio exceeding US$27 billion across 160+ projects; HNP ambition: quality, affordable health services to 1.5 billion people by 2030.

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APPENDIX A

# Glossary of Constructs

**Table A1. Defined terms**

| Term | Definition |
|---|---|
| OCCR | Open Constitutional Clinical Reasoning — L1 constitutional layer of CLAI-OS v2.0[1] |
| Deterministic-first chain | Four-node reasoning topology (evidence → framework → intermediate conclusions → decision); every transformation demonstrable, every node challengeable[1] |
| Constitutional constraint | Hard rule (population genetics, contraindications, epidemiology) that statistical optimization cannot override; early-termination halt on violation[1] |
| EPI | Equity Parity Index — max $\lvert PPV_i - PPV_j \rvert$ across deployed strata; mandated < 0.05; breach → adapter revision + explicit uncertainty[1] |
| Evidence coverage ρ | Ratio of required evidence present to required; below ρ* → output "indeterminate"; no imputation[1] |
| Contradiction latency CL | Time from contradictory-evidence insertion to explicit-uncertainty flag; ceiling 500 ms with node identification[1] |
| Override capture / rate | Every clinician override logged with justification, timestamp, ID in hash chain; rate > 15% triggers mandatory recalibration[1] |
| Resilience envelope RE | 72 h bandwidth-zero operation; solar-only grid-zero execution; deterministic merge under 30-day partition[1] |
| Sovereignty verification SV | Primary reasoning on sovereign-local infrastructure; signed adapter provenance; no foreign-cloud dependency[1] |
| CGCA | Constitutional-Grade Clinical AI — the five-gate conformance predicate of relation (5) (this blueprint, contribution 1) |
| EPI-linked disbursement | Results-based financing with measured parity as condition precedent (contribution 2) |
| TCD / TCS / t* | Total Cost of Dependency / Total Cost of Sovereignty / break-even year (contribution 3) |
| Adapter Registry | Signed, versioned, nationally peer-reviewed store of national compliance adapters[1] |
| Silent mode | Trial phase where chains run in parallel with standard care, outputs logged but not acted on |
| AEGIS-GLOBAL | Open-architecture technology-transfer framework enabling sovereign national manufacture[1] |

APPENDIX B

# Indicator and Verification Matrix

**Table B1. Programme indicators — every indicator has a source, a cadence and a hypothesis it serves**

| Indicator | Source instrument | Frequency | Hypothesis served |
|---|---|---|---|
| Sensitivity / serious-harm-equivalent error vs control | Silent-mode adjudication | Per case series | H1 |
| Indeterminate rate by domain | Chain telemetry | Continuous | H2 |
| Contradiction detection rate and latency | Falsifiability harness (seeded faults) | Per certification run | H3 |
| Envelope pass/fail (72 h, solar, 30-day partition) | Hardware-in-the-loop rigs | Per certification run | H4 |
| EPI by stratum; days-in-breach | EPI Observatory | Continuous; quarterly review | H5 |
| Ledger-consistency under red-team gaming | Adversarial audit programme | Semi-annual | H6 |
| Unit cost, transfer capex, break-even t* | Corridor cost ledger | Quarterly | H7 |
| Regulatory review duration (matched dossiers) | Regulator partnership records | Per dossier | H8 |
| Override capture completeness; override rate | Audit chain analytics | Continuous | Agency gate (all) |

APPENDIX C

# Adapter Deployment Profiles

The ten adapters of the parent taxonomy deploy in waves matched to pilot domains and national priorities. Profiles below define the sequencing frame; all adapters remain nationally governed under the Registry.[1]

**Table C1. Adapter deployment waves (frame, not forecast)**

| Wave | Adapters | Rationale |
|---|---|---|
| **Wave 1 — pilot-critical** | population genetics; data sovereignty; age-specific dosing | Pharmacogenomic dosing corridor (Pilot B) and sovereignty verification are the conformance backbone; CYP2C19/warfarin evidence base is mature[23],[24] |
| **Wave 2 — equity-critical** | environmental exposure; reproductive health; resistance pharmacology | Strata where population-bound failure is best documented; local resistance surveillance is a national asset[1] |
| **Wave 3 — system-critical** | ethical frameworks; mental-health liability; traditional medicine | National ethical and legal frameworks require longest consultation cycles; highest litigation sensitivity |
| **Standby — contingency** | conflict triage | Deployed only under explicit national humanitarian-law review and command authority[1] |

APPENDIX D

# Sensitivity Analysis

The thesis's exposure to its four most volatile parameters is bounded below. The stress envelope differs from the base case principally in timing: silent-mode outcomes, indeterminate rates, corridor break-even and regulatory recognition each shift the adoption curve, not its direction — because the programme's cost is small relative to the procurement and portfolio flows it addresses.

**Table D1. Wedge sensitivity (base case; illustrative)**

| Parameter shock | Effect on certification | Effect on disbursement | Effect on corridor |
|---|---|---|---|
| H1 narrow: non-inferiority in 8 of 15 diseases only | Domain-scoped marks issued | Unchanged (parity instrument independent) | Unchanged |
| Indeterminate rate at 20% (upper tolerance) | Gate 2 passes with clinical-review protocol | Unchanged | Unchanged |
| t* at 7 years (slow limit) | Unchanged | Unchanged | K4 boundary; milestone pricing restructured |
| Regulatory recognition delayed 24 months | Sandbox revenue shifts right; ministry certifications continue | Unchanged | Unchanged |
| Pilot funding −40% | Harness and silent-mode core unaffected (lab-scale) | Covenant pilot deferred to Phase 3 | Transfer package descoped to documentation |

The directional conclusion is robust: no single observed-variable shock closes the framework's case, because the four timing forces of Part I are independent. What would close it is stated in Table 11 — which is why the programme stages capital behind falsification gates and treats a clean rejection as a publishable, valuable result.[1]

APPENDIX E

# Illustrative Corridor Computation: Dependency versus Sovereignty

This appendix instantiates relation (6) with stylized, clearly labelled design-target parameters so that the break-even mechanics of contribution 3 are visible end to end. Base case, per country-programme, in US$ millions: dependency costs open at 2.60 per year and escalate at 10 percent annually on licence repricing and integration drag; transfer capex of 5.20 — fabrication, validation and workforce transfer — is charged in full at year 0; sovereign manufacture and national maintenance begin at 2.70 per year and decline 8 percent annually on the domestic learning curve; all flows are discounted at 8 percent. The slow-limit parameterization (2.50 opening, 9 percent growth, 6.20 capex, 2.90 operations, 5 percent learning) extends t* to 7 — the K4 boundary of Table 11. Every figure is illustrative — a planning skeleton to be replaced by the pilot-country cost ledger (Table 16; Appendix B) — not a finding; and the parameterization is deliberately conservative against the sovereignty path, which is assigned no value for employment, security of supply or strategic autonomy.

**Table E1. Worked computation, base case (US$ millions, stylized). Cumulative TCS crosses below cumulative TCD during year 4 — inside hypothesis H7's five-year bound.**

| Year | Dependency annual | Discounted | Cumulative TCD | Sovereignty annual | Discounted | Cumulative TCS |
|---|---|---|---|---|---|---|
| 0 | 2.60 | 2.60 | 2.60 | — (build year) | 0.00 | 5.20 |
| 1 | 2.86 | 2.65 | 5.25 | 2.70 | 2.50 | 7.70 |
| 2 | 3.15 | 2.70 | 7.95 | 2.48 | 2.13 | 9.83 |
| 3 | 3.46 | 2.75 | 10.69 | 2.29 | 1.81 | 11.64 |
| **4** | **3.81** | **2.80** | **13.49** | **2.10** | **1.55** | **13.19** |
| 5 | 4.19 | 2.85 | 16.34 | 1.93 | 1.32 | 14.51 |
| 6 | 4.61 | 2.90 | 19.24 | 1.78 | 1.12 | 15.63 |
| 7 | 5.07 | 2.96 | 22.20 | 1.64 | 0.96 | 16.58 |
| 8 | 5.57 | 3.01 | 25.21 | 1.51 | 0.81 | 17.40 |
| 9 | 6.13 | 3.07 | 28.28 | 1.39 | 0.69 | 18.09 |
| 10 | 6.74 | 3.12 | 31.40 | 1.27 | 0.59 | 18.68 |

Three properties carry the argument. First, **the crossing is early and robust**: t* = 4 in the base case, 4–5 across discount rates from 5 to 12 percent, and 7 only at the slow limit — precisely the K4 boundary (Table 11; Table D1). Second, **the dividend compounds**: by year 10 the cumulative gap is US$12.7 million per country-programme, widening at roughly US$5.5 million per year — value accruing to national accounts rather than to external vendors. Third, **the asymmetry is structural**: the dependency path's growth is set by a counterparty's pricing power; the sovereignty path's decline is set by domestic learning. The two curves are governed by different owners — which is the entire point of the corridor. H7 tests whether pilot-country cost structures reproduce t* ≤ 5; K4 closes or restructures the corridor honestly if they do not (Part VII).

The slow-limit computation is shown in full because it defines the corridor's honest failure mode: even with higher transfer capex, softer learning and slower licence escalation, the crossing arrives at year 7 with the dividend still compounding thereafter — the difference is a pricing and milestone problem, not a refutation of the path.

**Table E2. Worked computation, slow limit (US$ millions, stylized). Cumulative TCS crosses below cumulative TCD during year 7 — the K4 boundary case.**

| Year | Dependency annual | Discounted | Cumulative TCD | Sovereignty annual | Discounted | Cumulative TCS |
|---|---|---|---|---|---|---|
| 0 | 2.50 | 2.50 | 2.50 | — (build year) | 0.00 | 6.20 |
| 1 | 2.73 | 2.52 | 5.02 | 2.90 | 2.69 | 8.89 |
| 2 | 2.97 | 2.55 | 7.57 | 2.75 | 2.36 | 11.25 |
| 3 | 3.24 | 2.57 | 10.14 | 2.62 | 2.08 | 13.32 |
| 4 | 3.53 | 2.59 | 12.73 | 2.49 | 1.83 | 15.15 |
| 5 | 3.85 | 2.62 | 15.35 | 2.36 | 1.61 | 16.76 |
| 6 | 4.19 | 2.64 | 17.99 | 2.24 | 1.41 | 18.17 |
| **7** | **4.57** | **2.67** | **20.66** | **2.13** | **1.24** | **19.42** |
| 8 | 4.98 | 2.69 | 23.35 | 2.03 | 1.09 | 20.51 |
| 9 | 5.43 | 2.72 | 26.07 | 1.92 | 0.96 | 21.47 |
| 10 | 5.92 | 2.74 | 28.81 | 1.83 | 0.85 | 22.32 |

Read as a pair, Tables E1 and E2 bound the corridor's timing risk between years 4 and 7 on stylized parameters, with the direction of the sovereignty dividend unchanged across the envelope — the same directional robustness that Appendix D reports for the thesis as a whole. The pilot cost ledger replaces both tables with observed quantities; until then, every number on this page is a design target carrying a falsification handle, which is the standard this blueprint applies to itself.

> **Endnote on evidence discipline.** All quantitative claims in this blueprint trace to the cited sources as of October 2026. Where estimates carry ranges (e.g., diagnostic-harm plausible ranges[3]), the range is stated. Forward-looking quantities — pilot budgets, indeterminate tolerances, break-even years — are design targets to be replaced by metered figures under the pre-registered protocols of Part V, not findings. Nothing herein is medical advice, regulatory clearance, or a guarantee of clinical outcomes — which are, by design, the programme's to falsify.